

ECOCHAIN BLUE – TRUSTED DIGITAL MRV FOR BLUE CARBON ECOSYSTEMS

CSA0804 – PYTHON PROGRAMMING

MINI PROJECT

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INTRODUCTION

- ☐ **Blue carbon** refers to carbon stored in coastal and marine ecosystems.
- ☐ Current project data and carbon records can be **fragmented and difficult to verify**.
- ☐ Blockchain enables a **transparent and tamper-resistant registry**.
- ☐ The system records **project, monitoring, verification and carbon data** securely.
- ☐ Dashboards provide **real-time visualization, tracking and reporting**.



PROBLEM STATEMENT

- ☐ Lack of a **centralized and transparent** blue-carbon registry.
- ☐ Difficulty in **tracking carbon data and project progress**.
- ☐ Risk of **data manipulation or inconsistent records**.
- ☐ Limited transparency in **monitoring, reporting and verification (MRV)**.
- ☐ Need for a **secure, verifiable and accessible digital platform**.



LITERATURE SURVEY

Author(s) & Year	Title / Focus	Key Contribution	Limitation / Research Gap
Akhand et al. (2026)	Allochthonous particulate organic carbon in blue carbon ecosystems	Examines external organic carbon inputs and their impact on carbon accounting and stability .	Focuses mainly on carbon accounting; does not provide a digital blockchain-based MRV system .
Dinakarkumar et al. (2026)	Blue carbon sequestration and pollution control	Reviews mechanisms, processes and climate benefits of blue carbon ecosystems.	Primarily a review; lacks real-time monitoring, verification and transparent data management .
Wahsheh (2026)	Cryptographic framework for coastal ecosystem monitoring	Proposes cryptographic methods to improve the security of blue-carbon accounting and monitoring data.	Focuses on security; does not integrate a complete blockchain registry, dashboard and reporting system .
Maharani et al. (2025)	Transparent MRV in mangrove restoration	Presents a framework for transparent Monitoring, Reporting and Verification (MRV) of mangrove restoration.	Provides an MRV framework but lacks a comprehensive blockchain-based implementation and integrated registry .

OBJECTIVES

- Register and manage **blue-carbon coastal projects**.
- Monitor and calculate **carbon absorption**.
- Verify carbon data through **MRV processes**.
- Secure project records using **blockchain technology**.
- Generate **verified carbon credits** for stored carbon.

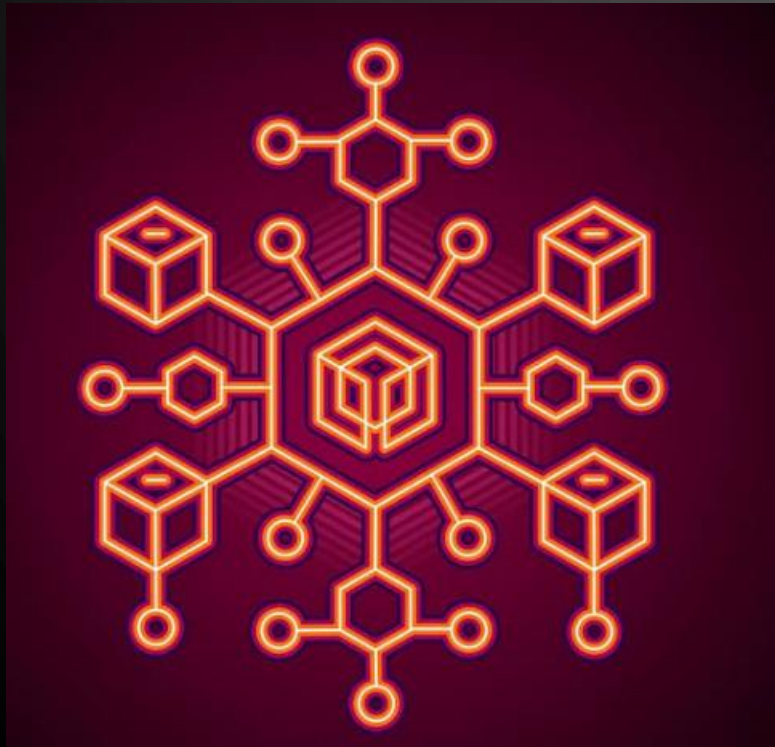


MODULE 1- PROJECT REGISTRATION & MONITORING



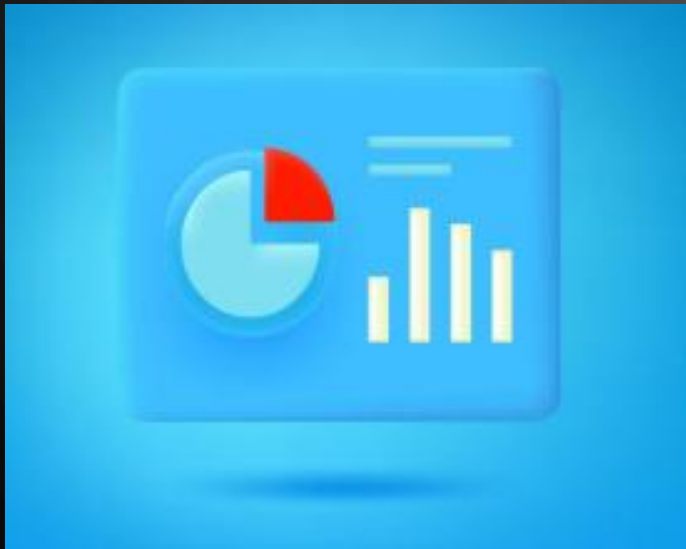
- Register **coastal projects and locations.**
- Store project and ecosystem details.
- Record **carbon absorption and monitoring data.**
- Track project progress through dashboards.

MODULE 2 - VERIFICATION & BLOCKCHAIN



- Validate submitted **carbon and monitoring data**.
- Maintain **transparent verification records**.
- Store verified information securely on **blockchain**.
- Provide an immutable **transaction history**.

MODULE 3 - CARBON CREDITS & REPORTING



- Calculate carbon storage eligible for credits.
- Issue **carbon credits after verification.**
- Display carbon-credit and project status.
- Generate **reports and visual analytics.**

SOVEREIGN REGISTRY & DIGITAL MRV

Global Blue Carbon Intelligence

Immutable on-chain accounting of coastal mangrove, seagrass, and salt marsh carbon sequestration sinks.

Register New Project

Submit MRV Telemetry

Registered Coastal Area



35,600 ha

↑ 100% Verified GIS Polygons

Baseline Carbon Stock



535,006 tCO₂e

IPCC Coastal Wetlands Tier 2/3

Verified Credits Issued



75,000 BLUE-C

• 1 Token = 1 Metric Ton CO₂e

Permanently Retired



2,500 tCO₂e

ESG Offsets Sealed on Chain

Sequestration Trajectory & IPCC Projection

Comparing Baseline Estimates against On-Chain Verified MRV Epochs
(tCO₂e)

Actual Verified
Baseline
Projected



Carbon Stock Composition

Aboveground vs Belowground vs Soil Organic Carbon



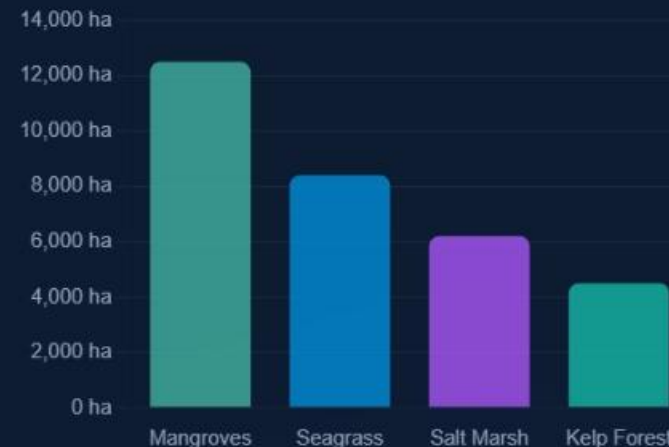
Soil Organic Carbon (SOC) Belowground Roots (BGB)
Aboveground Canopy (AGB)



High Soil Retention: Coastal Blue Carbon

Ecosystem Distribution

Active Restoration Hectares by Habitat



ADVANTAGES & LIMITATIONS

✓ Advantages

- ❖ **Transparent** and tamper-resistant records.
- ❖ Improves **trust in carbon data**.
- ❖ Simplifies **MRV and project tracking**.
- ❖ Enables secure **carbon-credit management**.
- ❖ Provides easy-to-understand **dashboards and reports**.

⚠ Disadvantages

- ❖ Blockchain implementation can be **complex**.
- ❖ Requires reliable **monitoring data and verification**.
- ❖ Initial development and maintenance may be **costly**.
- ❖ Blockchain transactions may involve **processing and scalability issues**.



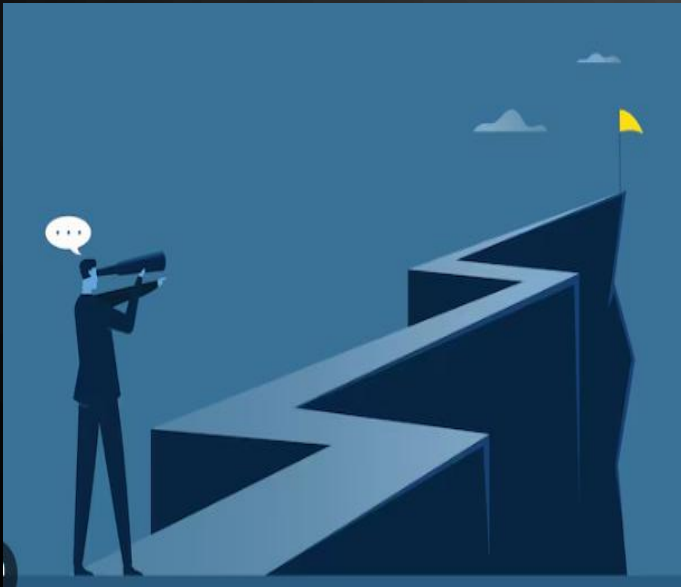
RESULTS & DISCUSSIONS

- ☐ Successfully registers and tracks **blue-carbon projects**.
- ☐ Provides centralized monitoring of **carbon absorption data**.
- ☐ Verified records are securely stored using **blockchain**.
- ☐ Dashboards enable easy **data visualization and tracking**.
- ☐ Verified carbon storage can be converted into **carbon credits**.





FUTURE SCOPE



- ☐ Integrate **IoT sensors and satellite data** for automated monitoring.
- ☐ Add **AI-based carbon estimation and anomaly detection**.
- ☐ Support **multiple blockchain networks**.
- ☐ Enable integration with **carbon-credit marketplaces**.
- ☐ Expand the system to support **large-scale coastal projects**.

THANK YOU